

Pfizer-BioNTech COVID-19 Vaccine

EUA 27034

Response to CBER 27 November 2020 Information Request

November 2020

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1. INTRODUCTION

Reference is made to Request for Emergency Use Authorization (EUA 27034) submitted 20 November 2020 by Pfizer on behalf of BioNTech. Further reference is made to the clinical information amendment submitted to EUA 27034 on 25 November 2020 (Sequence 0004) and CBER's Information Request regarding this amendment communicated by email from (b) (6), OVRP, CBER to Elisa Harkins, Pfizer Inc. on 17 November 2020.

CBER questions are presented in ***bold italics*** and Pfizer/BioNTech's responses follow in plain text.

2. CBER QUESTIONS FROM 27 NOVEMBER 2020

2.1. CBER Question 1

Regarding tables 26 and 27 from the 508 compliant tables document you submitted in Microsoft Word format: Table 26: Percentage of subjects reporting the occurrence SAEs from Dose 1 to 1 Month After Dose 2 classified by MedDRA Primary System Organ Class and Preferred Term – Safety Population* and Table 27: Percentage of subjects reporting the occurrence SAEs from Dose 1 to 1 Month After Dose 2 classified by MedDRA Primary System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status - Safety Population* Baseline SARS-CoV-2 Status: Positive. Please clarify the cutoff date for the data summarized in these tables. Please also clarify if the follow-up period is one month follow-up post dose 2 or is it up to the November 14, 2020, cutoff date. We had requested all safety information available up to the November 14, 2020, cutoff. Please confirm and comment.

Response

The data cutoff date for the EUA snapshot and analyses was 14 November 2020. SAEs reported by the cutoff date and within the 1-month post-Dose 2 follow-up period were included in Tables 26 and 27 of the 508-compliant tables document.

2.2. CBER Question 2

Regarding the 58 subjects who received a higher dose due to a medication error, please provide us with a listing of these subjects and indicate whether we can identify them with any variable in the database. We cannot identify a 50 µg dose in EX or ADSL variables. The ADAE we have AE Category Medication error, but the Incorrect dosages do not specify dose actually given. Could you please guide us to where we can find the variable or if there is a flag to use to identify these subjects?

Response

There were 52 subjects enrolled at study site #1231 in Argentina who received a higher dose (~58 µg) due to a medication error. These subjects can be identified in SDTM DV domain

using the following condition: where scan(USUBJID, 2, ' ') = '1231' and DVTERM = 'Dosing/administration error, subject did not receive correct dose of vaccine' and DVCAT = 'Important'.

2.3. CBER Question 3

Please provide a summary table by study arm of all participants who had a potential COVID-19 illness that was not confirmed to be COVID-19. Please include numbers and percentages of subjects in each study arm who had a potential COVID-19 illness at any time during follow-up, during the 7 days post-dose 1, and during the 7 days post-dose 2. Please include a listing of all these participants by study arm as well.

Response

Summary tables of participants with COVID-19 symptoms who were not confirmed by PCR to be a COVID-19 case at any time throughout follow-up after Dose 1 (Table 1), during the 7 days post Dose 1 (Table 2), or during the 7 days post Dose 2 (Table 3), are presented below. A listing of these participants is included in Module 5.3.5.1.

Table 1. Subjects With Symptoms of COVID-19 but not Confirmed Cases by PCR – Dose 1 All-Available Efficacy Population

	Vaccine Group (as Randomized)		Total n ^a (%)
	BNT162b2 (30 µg) n ^a (%)	Placebo n ^a (%)	
Subjects developed protocol defined symptoms after dose 1 but not confirmed by PCR ^b	1594	1816	3410
PCR Results Negative	1252 (78.5)	1416 (78.0)	2668 (78.2)
PCR Results unavailable or unknown	342 (21.5)	400 (22.0)	742 (21.8)
Swab not taken	226 (14.2)	231 (12.7)	457 (13.4)
Swab taken but results not available	80 (5.0)	132 (7.3)	212 (6.2)
Swab taken outside of symptom window	36 (2.3)	37 (2.0)	73 (2.1)
a. n = Number of subjects with the specified characteristic. b. The values in this row are the denominators for the percentage calculations. PFIZER CONFIDENTIAL SDTM Creation: 17NOV2020 (09:54) Source Data: adsympt Table Generation: 25NOV2020 (14:44) (Cutoff Date: 14NOV2020, Snapshot Date: 16NOV2020) Output File: ./nda2_unblinded/C4591001_EUA_FAEF_RR/adsympt_sympt_pcr_d1_aai			

Table 2. Subjects With Symptoms of COVID-19 but not Confirmed Cases by PCR During 7 Days After Dose 1 – Dose 1 All-Available Efficacy Population

	Vaccine Group (as Randomized)		
	BNT162b2 (30 µg)	Placebo	Total
	n ^a (%)	n ^a (%)	n ^a (%)
Subjects developed protocol defined symptoms during 7 days after Dose 1 but not confirmed by PCR ^b	168	185	353
PCR Results Negative	111 (66.1)	123 (66.5)	234 (66.3)
PCR Results unavailable or unknown	57 (33.9)	62 (33.5)	119 (33.7)
Swab not taken	43 (25.6)	50 (27.0)	93 (26.3)
Swab taken but results not available	8 (4.8)	6 (3.2)	14 (4.0)
Swab taken outside of symptom window	6 (3.6)	6 (3.2)	12 (3.4)
a. n = Number of subjects with the specified characteristic. b. The values in this row are the denominators for the percentage calculations. PFIZER CONFIDENTIAL SDTM Creation: 17NOV2020 (09:54) Source Data: adsympt Table Generation: 28NOV2020 (15:21) (Cutoff Date: 14NOV2020, Snapshot Date: 16NOV2020) Output File: .nda2_unblinded/C4591001_EUA_FAEF_RR/adsympt_sympt_pcr_7dur_d1_aai			

Table 3. Subjects With Symptoms of COVID-19 but not Confirmed Cases by PCR During 7 Days After Dose 2 – Dose 2 All-Available Efficacy Population

	Vaccine Group (as Randomized)		
	BNT162b2 (30 µg)	Placebo	Total
	n ^a (%)	n ^a (%)	n ^a (%)
Subjects developed protocol defined symptoms during 7 days after Dose 2 but not confirmed by PCR ^b	241	164	405
PCR Results Negative	157 (65.1)	114 (69.5)	271 (66.9)
PCR Results unavailable or unknown	84 (34.9)	50 (30.5)	134 (33.1)

	Vaccine Group (as Randomized)		
	BNT162b2 (30 µg)	Placebo	Total
	n ^a (%)	n ^a (%)	n ^a (%)
Swab not taken	72 (29.9)	40 (24.4)	112 (27.7)
Swab taken but results not available	5 (2.1)	6 (3.7)	11 (2.7)
Swab taken outside of symptom window	7 (2.9)	4 (2.4)	11 (2.7)

a. n = Number of subjects with the specified characteristic.
b. The values in this row are the denominators for the percentage calculations.
PFIZER CONFIDENTIAL SDTM Creation: 17NOV2020 (09:54) Source Data: adsympt Table Generation: 28NOV2020 (15:27)
(Cutoff Date: 14NOV2020, Snapshot Date: 16NOV2020) Output File: ./nda2_unblinded/C4591001_EUA_FAEF_RR/adsympt_sympt_pcr_7dur_d2_aai

2.4. CBER Question 4

Regarding the efficacy outcomes stratified by risk factors, with the purpose of obtaining more COVID-19 specific risks and meaningful information, we request that you provide analyses by each of the following risk factors for COVID-19 disease. We had agreed on reporting efficacy outcomes by obesity as a risk factor independently of others (35% of the population). You may also report efficacy outcomes by number of subjects who have 1 or more comorbidities that increase the risk of severe COVID-19 disease: defined as patients who meet at least one of the following criteria:

- *≥65 yo*
- *Chronic lung disease (e.g., emphysema and chronic bronchitis, idiopathic pulmonary fibrosis, and cystic fibrosis) or moderate to severe asthma*
- *Significant cardiac disease (e.g., heart failure, coronary artery disease, congenital heart disease, cardiomyopathies, and pulmonary hypertension)*
- *Hypertension*
- *Obesity (body mass index ≥ 30 kg/m²)*
- *Diabetes (Type 1, Type 2 or gestational)*

Response

Vaccine efficacy for subjects ≥65 years of age was provided in the EUA submission. Efficacy analyses for the other individual risk factors requested above are presented in [Table 4](#) and [Table 5](#).

Table 4. Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2, by Comorbidity Status – Subjects Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

Efficacy Endpoint Subgroup	Vaccine Group (as Randomized)					
	BNT162b2 (30 µg) (N ^a =18198)			Placebo (N ^a =18325)		
	n1 ^b	Surveillance Time ^c (n2 ^d)	n1 ^b	Surveillance Time ^c (n2 ^d)	VE (%)	(95% CI ^e)
First COVID-19 occurrence from 7 days after Dose 2						
Overall	8	2.214 (17411)	162	2.222 (17511)	95.0	(90.0, 97.9)
Comorbidity						
No comorbidity	4	1.189 (9381)	76	1.197 (9482)	94.7	(85.9, 98.6)
Any comorbidity ^f	4	1.025 (8030)	86	1.025 (8029)	95.3	(87.7, 98.8)
Any malignancy	1	0.092 (704)	4	0.090 (681)	75.7	(-145.8, 99.5)
Cardiovascular	0	0.067 (534)	5	0.062 (492)	100.0	(-0.8, 100.0)
Chronic pulmonary disease	1	0.175 (1374)	14	0.171 (1358)	93.0	(54.1, 99.8)
Diabetes	1	0.176 (1372)	19	0.176 (1374)	94.7	(66.8, 99.9)
Obese (≥30.0 kg/m ²)	3	0.763 (6000)	67	0.782 (6103)	95.4	(86.0, 99.1)
Hypertension	2	0.567 (4413)	44	0.567 (4437)	95.4	(82.6, 99.5)
Diabetes (including gestational diabetes)	1	0.177 (1381)	20	0.178 (1384)	95.0	(68.7, 99.9)

Abbreviations: N-binding = SARS-CoV-2 nucleoprotein-binding; NAAT = nucleic acid amplification test; SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2; VE = vaccine efficacy.

Note: Subjects who had no serological or virological evidence (prior to 7 days after receipt of the last dose) of past SARS-CoV-2 infection (ie, N-binding antibody [serum] negative at Visit 1 and SARS-CoV-2 not detected by NAAT [nasal swab] at Visits 1 and 2), and had negative NAAT (nasal swab) at any unscheduled visit prior to 7 days after Dose 2 were included in the analysis.

a. N = number of subjects in the specified group.

b. n1 = Number of subjects meeting the endpoint definition.

c. Total surveillance time in 1000 person-years for the given endpoint across all subjects within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after Dose 2 to the end of the surveillance period.

d. n2 = Number of subjects at risk for the endpoint.

e. Confidence interval (CI) for VE is derived based on the Clopper and Pearson method adjusted for surveillance time.

f. Subject who had 1 or more comorbidities that increase the risk of severe COVID-19 disease: defined as subjects who had at least one of the Charlson comorbidity index category or BMI ≥30 kg/m².

PFIZER CONFIDENTIAL SDTM Creation: 17NOV2020 (09:54) Source Data: adc19ef Table Generation: 29NOV2020 (11:57)

Efficacy Endpoint Subgroup	Vaccine Group (as Randomized)				VE (%)	(95% CI ^e)
	BNT162b2 (30 µg) (N ^a =18198)		Placebo (N ^a =18325)			
	n1 ^b	Surveillance Time ^c (n2 ^d)	n1 ^b	Surveillance Time ^c (n2 ^d)		
(Cutoff Date: 14NOV2020, Snapshot Date: 16NOV2020) Output File: ./nda2_unblinded/C4591001_EUA_FAEF_RR/adc19ef_ve_cov_7pd2_wo_cg_eval						

Table 5. Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2, By Comorbidity Status – Subjects With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

Efficacy Endpoint Subgroup	Vaccine Group (as Randomized)				VE (%)	(95% CI ^e)
	BNT162b2 (30 µg) (N ^a =19965)		Placebo (N ^a =20172)			
	n1 ^b	Surveillance Time ^c (n2 ^d)	n1 ^b	Surveillance Time ^c (n2 ^d)		
First COVID-19 occurrence from 7 days after Dose 2						
Overall	9	2.332 (18559)	169	2.345 (18708)	94.6	(89.6, 97.6)
Comorbidity						
No comorbidity	5	1.250 (9975)	82	1.261 (10099)	93.8	(85.0, 98.1)
Any comorbidity ^f	4	1.083 (8584)	87	1.084 (8609)	95.4	(87.8, 98.8)
Any malignancy	1	0.096 (738)	4	0.092 (706)	76.0	(-142.3, 99.5)
Cardiovascular	0	0.070 (561)	5	0.065 (517)	100.0	(-0.8, 100.0)
Chronic pulmonary disease	1	0.184 (1463)	14	0.180 (1449)	93.0	(54.0, 99.8)
Diabetes	1	0.184 (1458)	19	0.186 (1470)	94.7	(66.6, 99.9)
Obese (≥30.0 kg/m ²)	3	0.810 (6445)	68	0.832 (6582)	95.5	(86.2, 99.1)
Hypertension	2	0.594 (4681)	44	0.595 (4714)	95.4	(82.6, 99.5)
Diabetes (including gestational diabetes)	1	0.186 (1467)	20	0.187 (1480)	95.0	(68.5, 99.9)
Abbreviation: VE = vaccine efficacy. a. N = number of subjects in the specified group.						

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Vaccine Group (as Randomized)						
Efficacy Endpoint Subgroup	BNT162b2 (30 µg) (N ^a =19965)		Placebo (N ^a =20172)		VE (%)	(95% CI ^e)
	Surveillance		Surveillance			
	n1 ^b	Time ^c (n2 ^d)	n1 ^b	Time ^c (n2 ^d)		
b. n1 = Number of subjects meeting the endpoint definition.						
c. Total surveillance time in 1000 person-years for the given endpoint across all subjects within each group at risk for the endpoint. Time period for COVID-19 case accrual is from 7 days after Dose 2 to the end of the surveillance period.						
d. n2 = Number of subjects at risk for the endpoint.						
e. Confidence interval (CI) for VE is derived based on the Clopper and Pearson method adjusted for surveillance time.						
f. Subject who had 1 or more comorbidities that increase the risk of severe COVID-19 disease: defined as subjects who had at least one of the Charlson comorbidity index category or BMI ≥30 kg/m ² .						
PFIZER CONFIDENTIAL SDTM Creation: 17NOV2020 (09:54) Source Data: adc19ef Table Generation: 29NOV2020 (11:55)						
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.nda2_unblinded/C4591001_EUA_FAEF_RR/adc19ef_ve_cov_7pd2_cg_eval						

2.5. CBER Question 5

Please clarify whether the ~8,000 subjects in the reactogenicity subset are indeed the first ~8,000 enrolled subjects. In the protocol, the plan was to obtain reactogenicity data from the first 6,000 subjects enrolled and included the safety population. Please confirm.

Response

In addition to the first ~6,000 subjects enrolled in Phase 2/3, the study plan was to also collect reactogenicity data for ~500 subjects in each country. The ~8,000 subjects in the reactogenicity subset in the EUA analyses included the first 6557 subjects enrolled in the US, and an additional 96 subjects from the US, the first 500 subjects enrolled in Argentina, the first 526 subjects enrolled in Brazil and the first 504 subjects enrolled in South Africa.

Please note that at the time of the BLA submission, the reactogenicity subset will be expanded to include ~500 subjects each from Germany and Turkey, as well as a subset in subjects with HIV, HCV, and HBV, subjects who are 16-17 years of age, and from younger subjects, 12-15 years of age.

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